

Code:

```
class Node:
```

```
    def __init__(self, data):
```

```
        self.data = data
```

```
        self.next = None
```

```
class Queue:
```

```
    def __init__(self):
```

```
        self.front = None
```

```
        self.rear = None
```

```
    def enqueue(self, data):
```

```
        new_node = Node(data)
```

```
        if self.rear is None:
```

```
            self.front = self.rear = new_node
```

```
        else:
```

```
            self.rear.next = new_node
```

```
            self.rear = new_node
```

```
        print(f"Enqueued {data}")
```

```
    def dequeue(self):
```

```
        if self.front is None:
```

```
            print("Queue Underflow")
```

```
            return None
```

```
        removed = self.front.data
```

```
        self.front = self.front.next
```

```
if self.front is None:
    self.rear = None
print(f"Dequeued {removed}")
return removed
```

```
def peek(self):
    if self.front is None:
        print("Queue is empty")
        return None
    print(f"Front element: {self.front.data}")
    return self.front.data
```

```
def display(self):
    if self.front is None:
        print("Queue is empty")
        return
    print("Queue (linked list):", end=' ')
    current = self.front
    while current:
        print(current.data, end=' ')
        current = current.next
    print()
```

```
queue = Queue()
```

```
queue.enqueue(10)
```

```
queue.enqueue(20)
```

```
queue.enqueue(30)
```

```
queue.display()
```

```
queue.dequeue()
```

```
queue.dequeue()
```

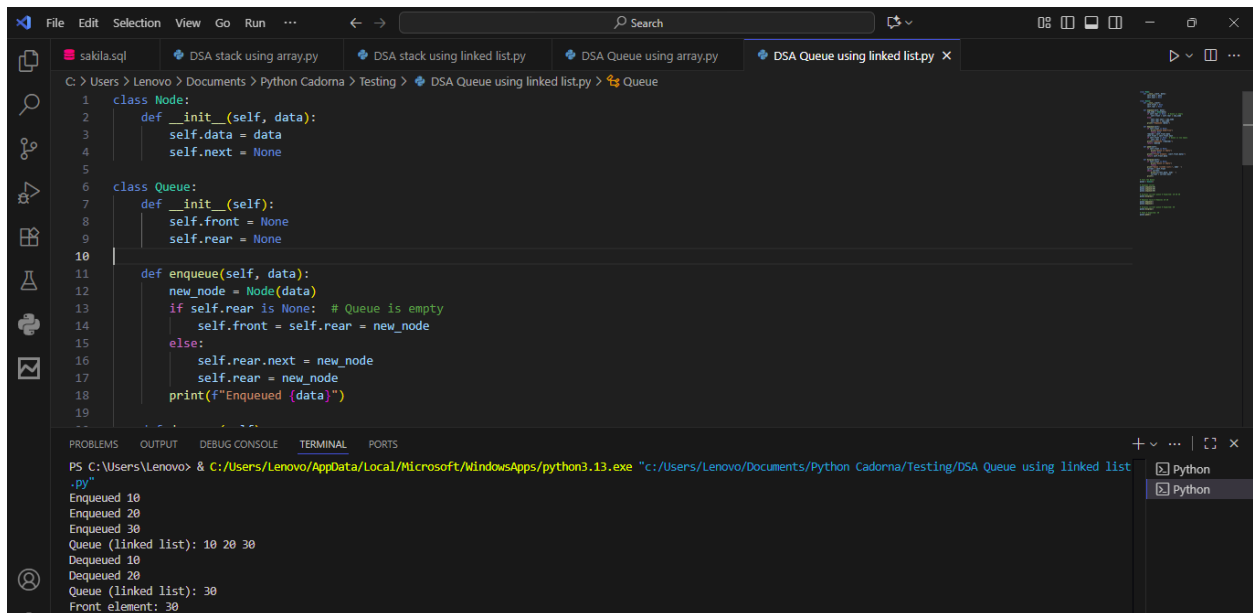
```
queue.display()
```

```
queue.peek()
```

Programiz Link:

<https://www.programiz.com/online-compiler/6bPOKZo8Jx2de>

Sample Output:



```
1 class Node:
2     def __init__(self, data):
3         self.data = data
4         self.next = None
5
6 class Queue:
7     def __init__(self):
8         self.front = None
9         self.rear = None
10
11     def enqueue(self, data):
12         new_node = Node(data)
13         if self.rear is None: # Queue is empty
14             self.front = self.rear = new_node
15         else:
16             self.rear.next = new_node
17             self.rear = new_node
18         print(f"Enqueued {data}")
19
20 if __name__ == '__main__':
21     q = Queue()
22     q.enqueue(10)
23     q.enqueue(20)
24     q.enqueue(30)
25     q.display()
26     q.dequeue()
27     q.dequeue()
28     q.display()
29     q.peek()
```

PS C:\Users\Lenovo> & c:\Users\Lenovo\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/Lenovo/Documents/Python Cadorna/Testing/DSA Queue using linked list.py"

Enqueued 10
Enqueued 20
Enqueued 30
Queue (linked list): 10 20 30
Dequeued 10
Dequeued 20
Queue (linked list): 30
Front element: 30